SOC Monitoring & Incident Response Lab

Wazuh — Complete Record of All Completed Scenarios

Portfolio-ready investigation report with evidence screenshots

Platform	Wazuh 4.14.7
Windows endpoint	Windows 11 — YASIN2 / yasin2
Wazuh Agent	ID 002 — yasin2
Wazuh Server	192.168.1.104
Completed scenarios	Scenario 1, Scenario 2, Scenario 3

Prepared from the controlled lab work and screenshots completed during the project.

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1. Project Overview

This project demonstrates a small Security Operations Centre (SOC) monitoring workflow using Wazuh as the SIEM/security monitoring platform and a Windows 11 virtual machine as the monitored endpoint. Three controlled scenarios were completed. Each scenario followed the same investigation pattern: generate a controlled activity, identify the Windows security event, allow Wazuh to collect and alert on it, investigate the evidence, classify the activity, and document the response.

Completed scenarios

Scenario	Activity tested	Windows event	Outcome
1	Single failed authentication	4625	Detected and investigated; controlled benign lab activity.
2	Repeated failed authentication	4625	Four repeated failures observed; controlled lab activity. No dedicated higher-level brute-force correlation rule was observed.
3	Process creation	4688	Detected by built-in rule 67027 and custom rule 100100; investigated as benign/expected Windows activity.

The report deliberately distinguishes detection from maliciousness. A security alert is an observation that requires investigation; it is not, by itself, proof of compromise.

2. Lab Architecture and Environment

Component	Configuration
Physical host	Windows laptop running VirtualBox
Wazuh Server VM	Wazuh 4.14.7 OVA; Manager + Indexer + Dashboard
Wazuh Server IP	192.168.1.104
Windows endpoint VM	Windows 11 Pro
Windows hostname	YASIN2
Wazuh agent name	yasin2
Wazuh agent ID	002
Windows endpoint IP	192.168.1.22
Agent transport	TCP 1514
Security log collection	Windows Security EventChannel

Connectivity was verified during Scenario 3 with Test-NetConnection to 192.168.1.104 on TCP port 1514, which returned TcpTestSucceeded: True. The Windows Wazuh service was also verified as Running.

3. Scenario 1 — Single Failed Authentication

Objective: demonstrate detection and investigation of a single failed Windows authentication attempt using Wazuh.

3.1 Controlled test

The Windows 11 VM was locked. One incorrect password was deliberately entered, followed by the correct password to regain access. This produced Windows Security Event ID 4625.

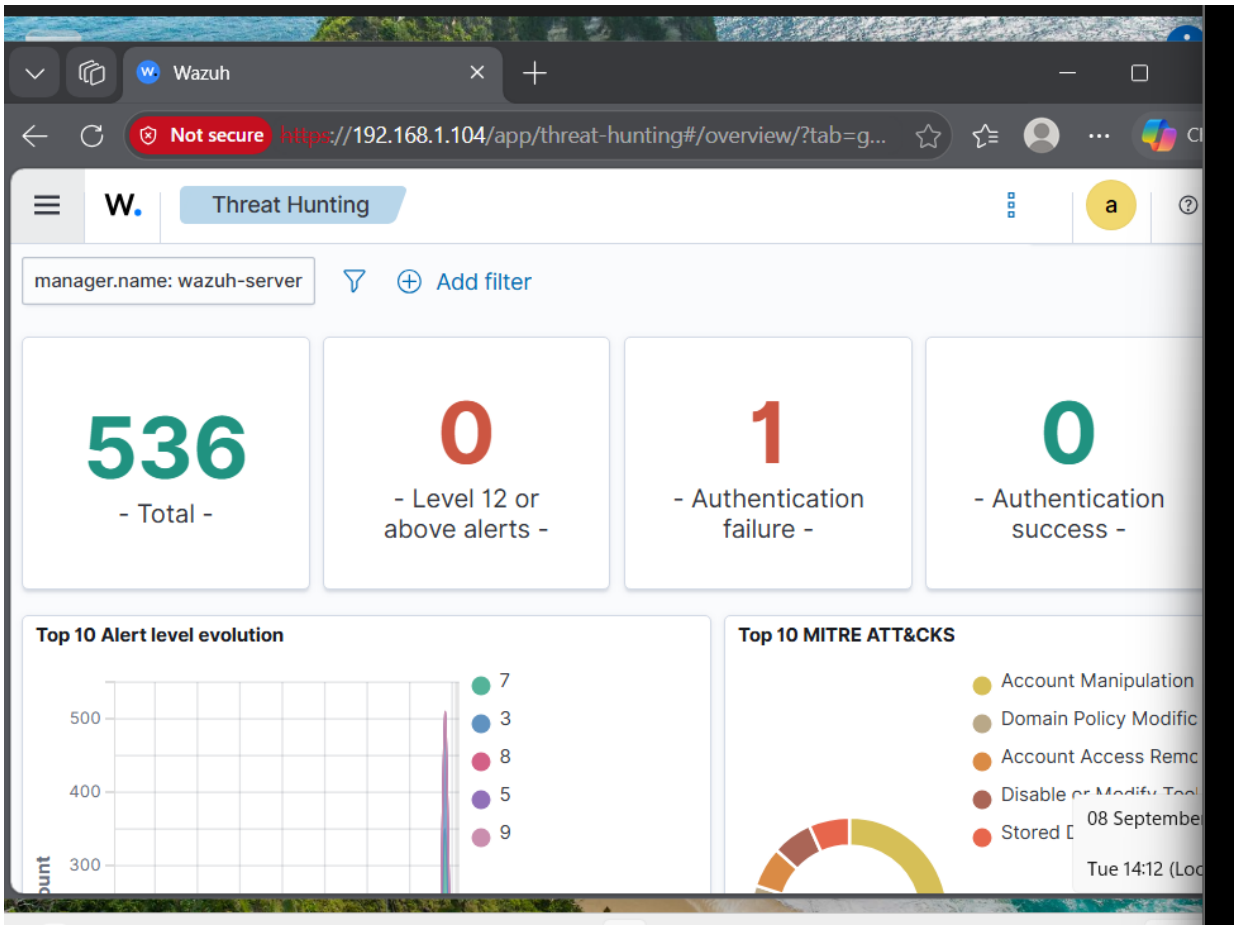


Figure 1. Wazuh Threat Hunting dashboard showing the authentication-failure alert.

3.2 Endpoint identification

The investigation was narrowed to the correct Windows endpoint. Agent 002 corresponds to the yasin2 endpoint.

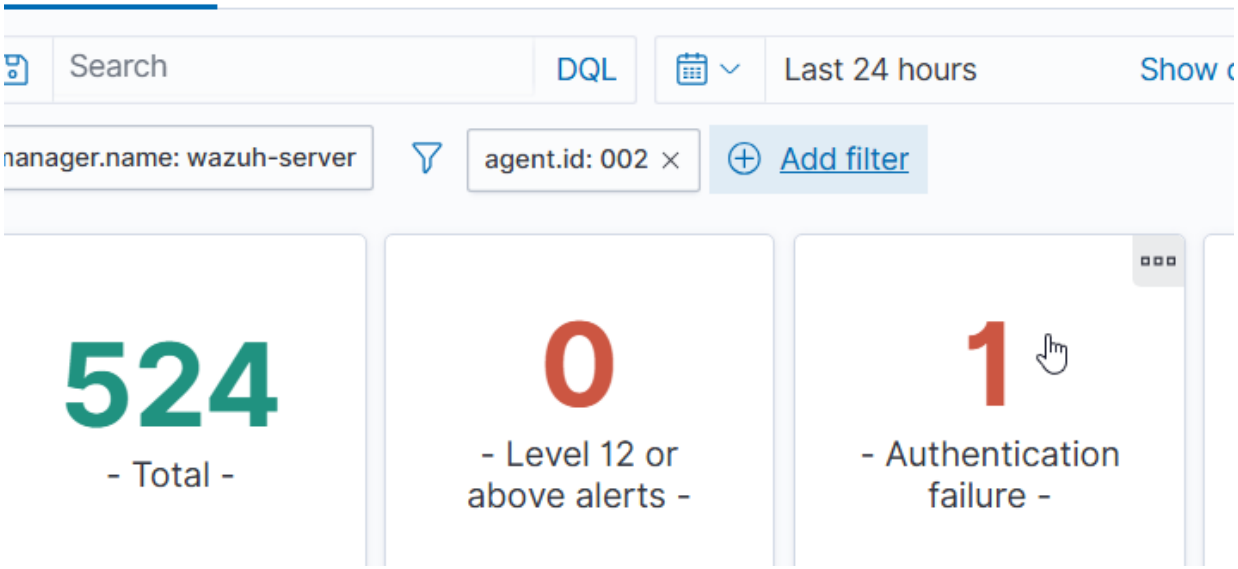


Figure 2. Wazuh filtering showing manager wazuh-server and Agent ID 002.

3.3 Finding the failed-logon event

The Events view was used to isolate the failed authentication event. One matching result was identified for yasin2 in the selected time period.

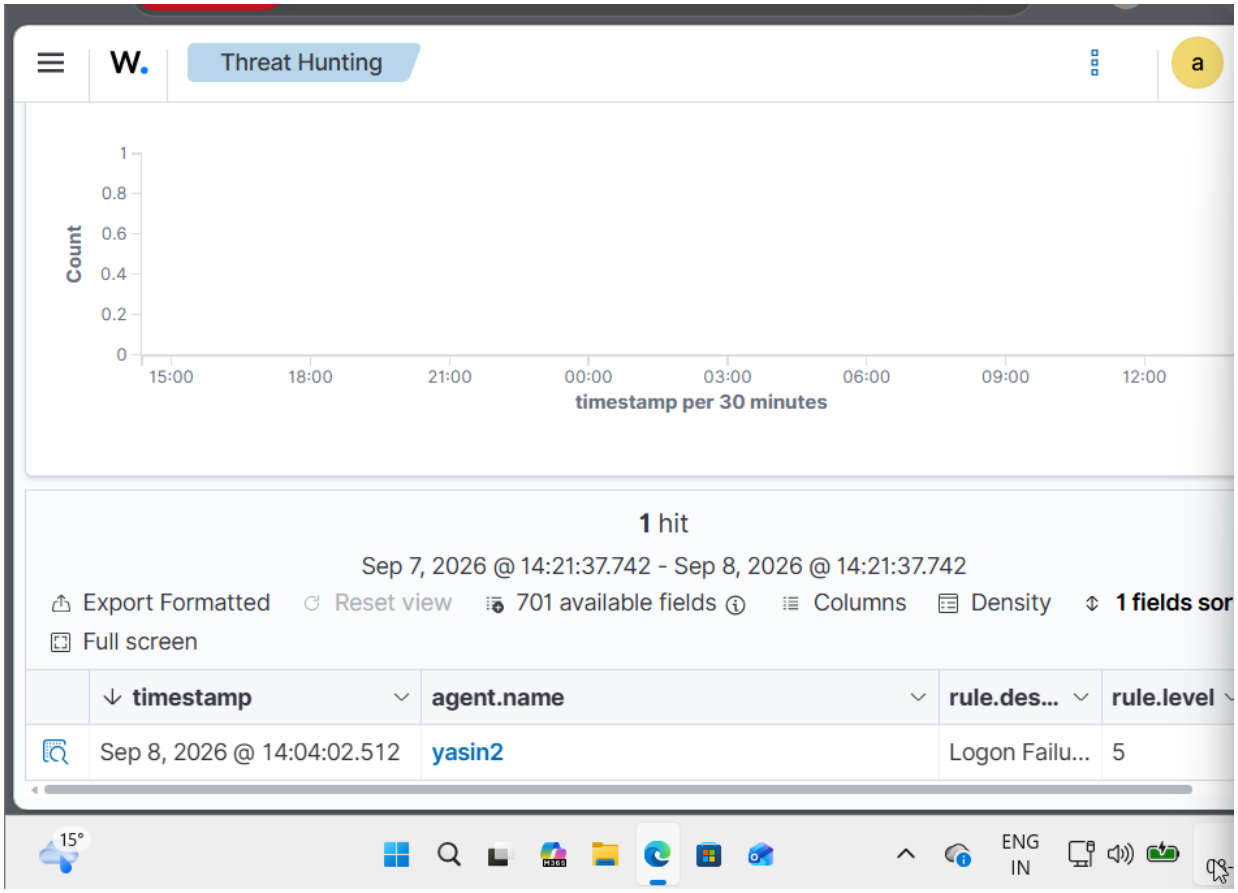


Figure 3. Events view showing one failed-logon result for yasin2.

3.4 Investigating the event details

The Wazuh event details were opened to inspect the authentication package, source address, logon process, logon type, status codes, and endpoint identity.

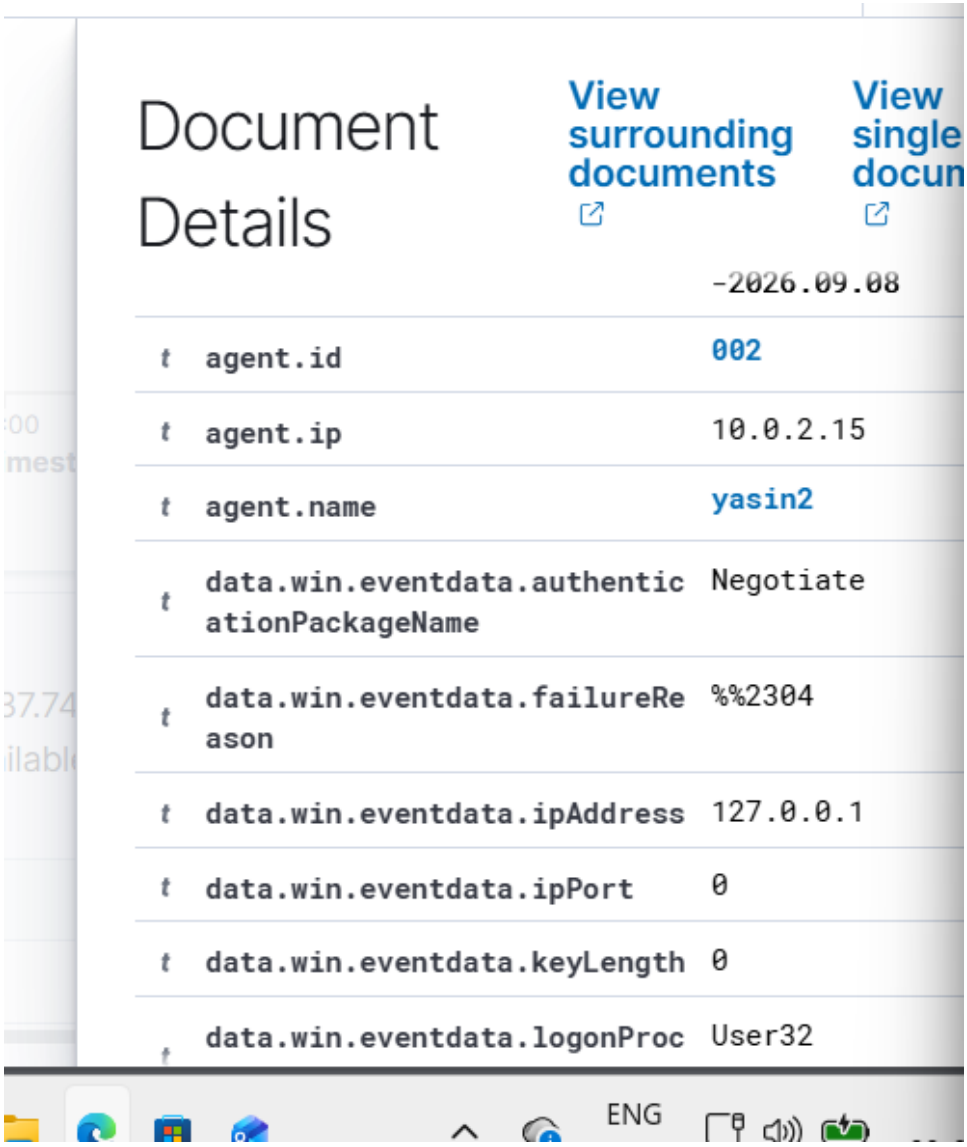


Figure 4. Event details showing Agent 002, yasin2, source address and authentication information.

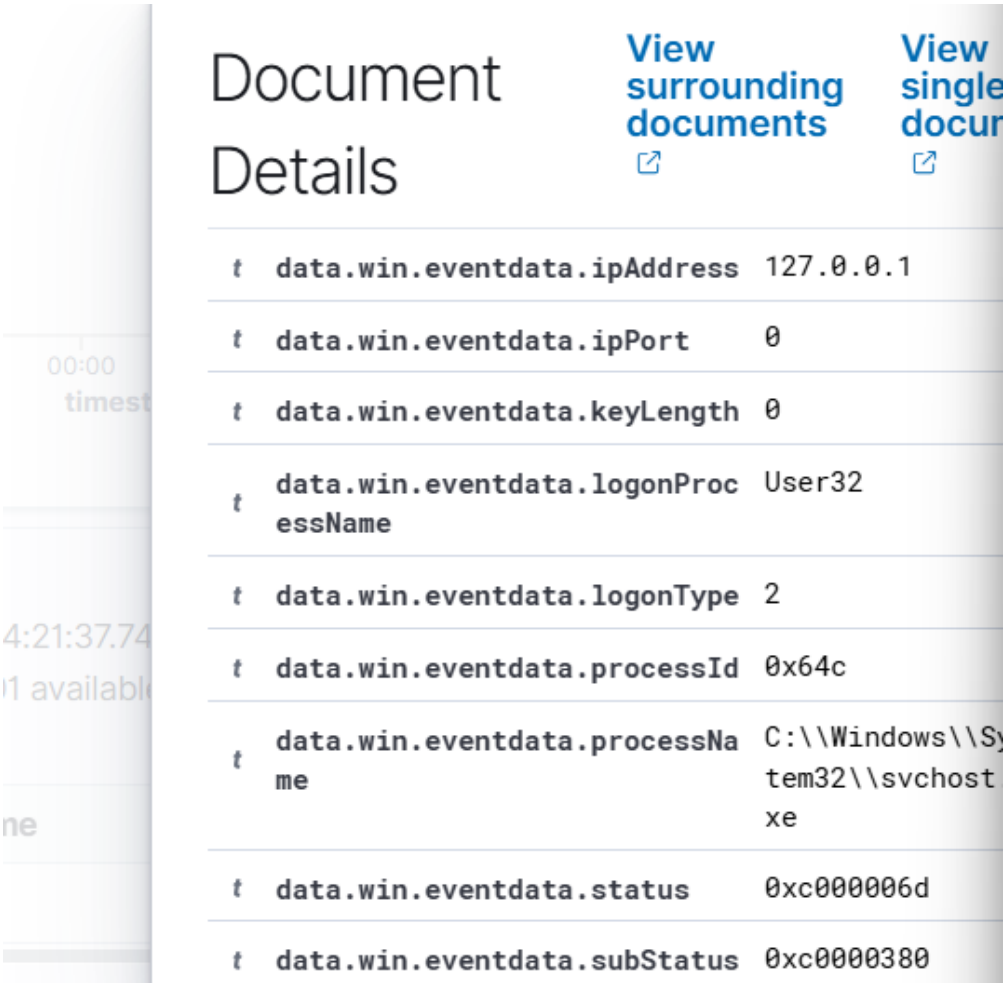


Figure 5. Event details showing Logon Type 2, process information and Windows status codes.

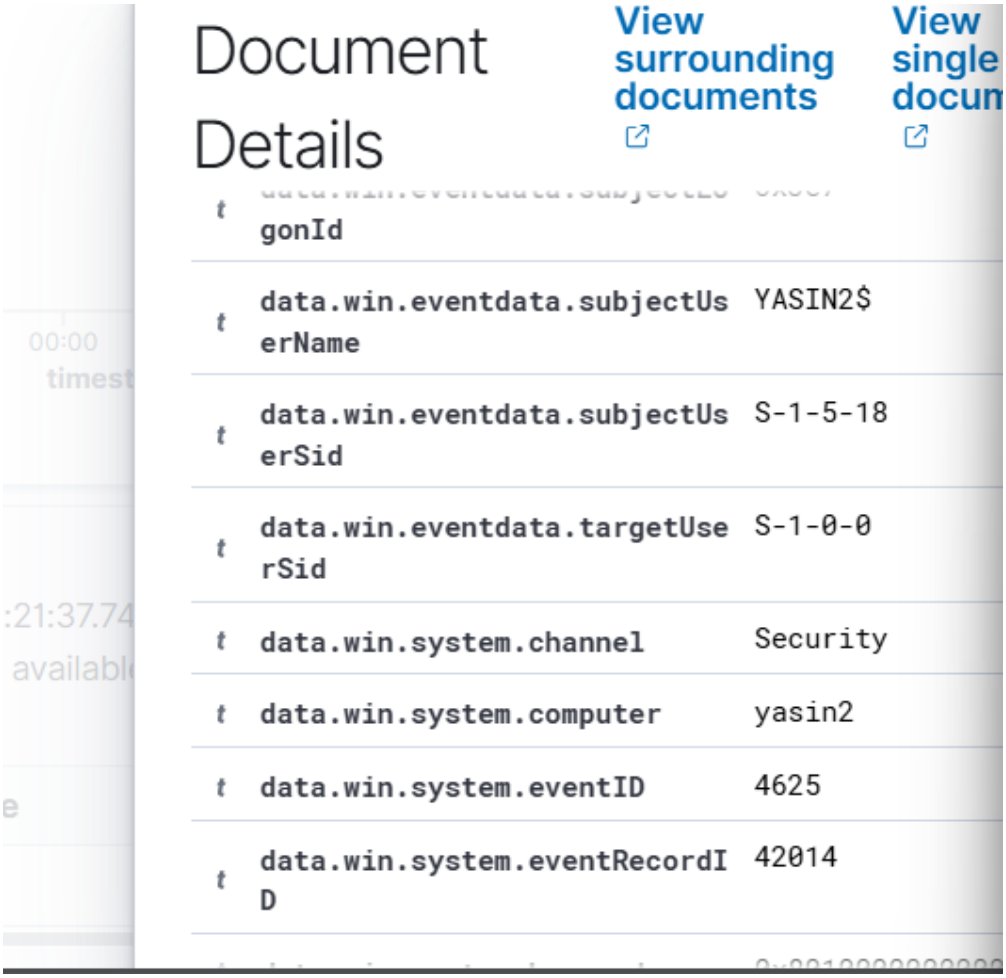


Figure 6. Event details confirming Windows Security Event ID 4625.

3.5 Classification and response

Item	Finding
Event ID	4625 — failed Windows logon
Agent	002 / yasin2
Logon type	2 (interactive)
Source IP	127.0.0.1 in the observed event
Process	C:\Windows\System32\svchost.exe
Status	0xc000006d
Classification	Controlled benign lab activity
Response	No containment required; activity was deliberately generated for testing.

3.6 Additional Event 4738

A nearby Event ID 4738 ('User account changed') was also observed for the account agsar. The event was treated as separate activity requiring verification. The investigation did not establish that it was caused by the 4625 event, and no unsupported causal claim is made.

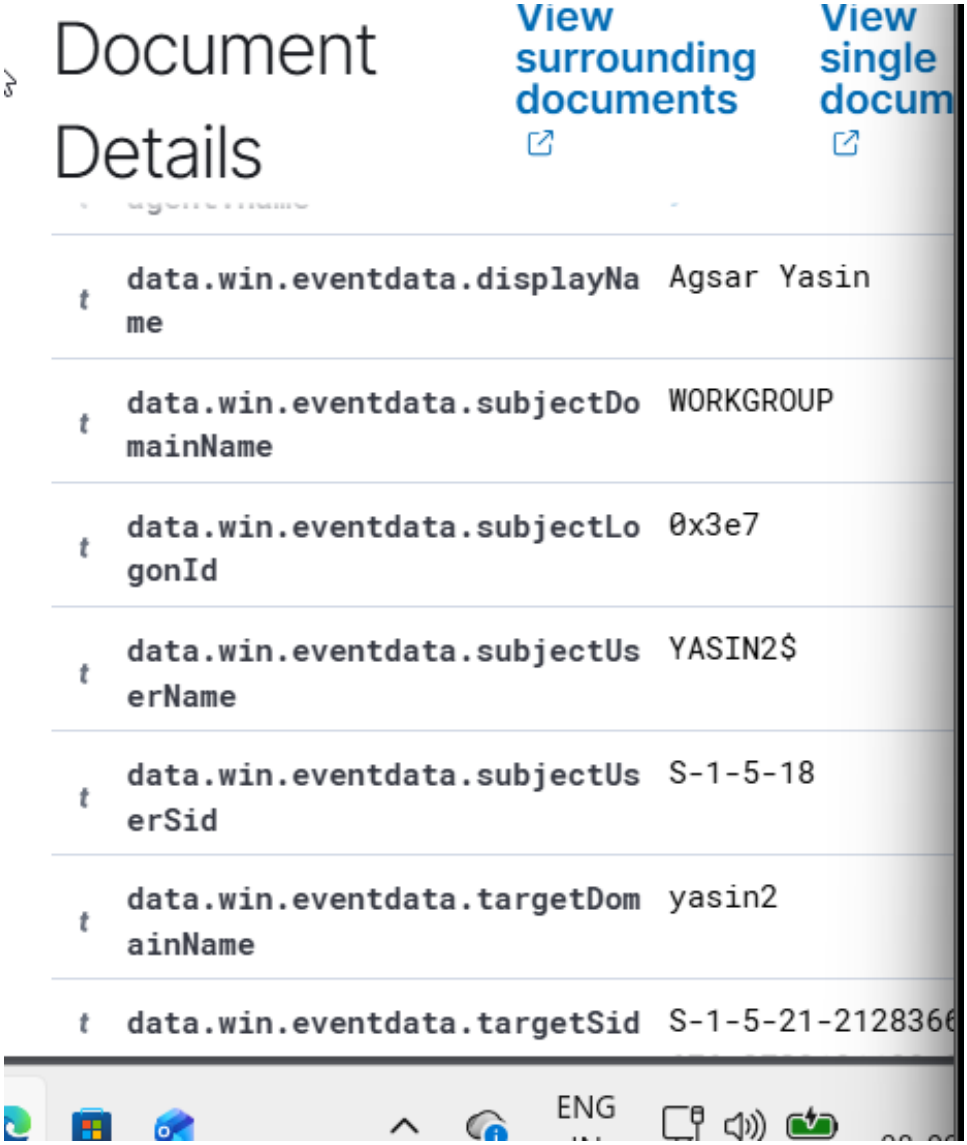


Figure 7. Event ID 4738 details identifying the target account as agsar.

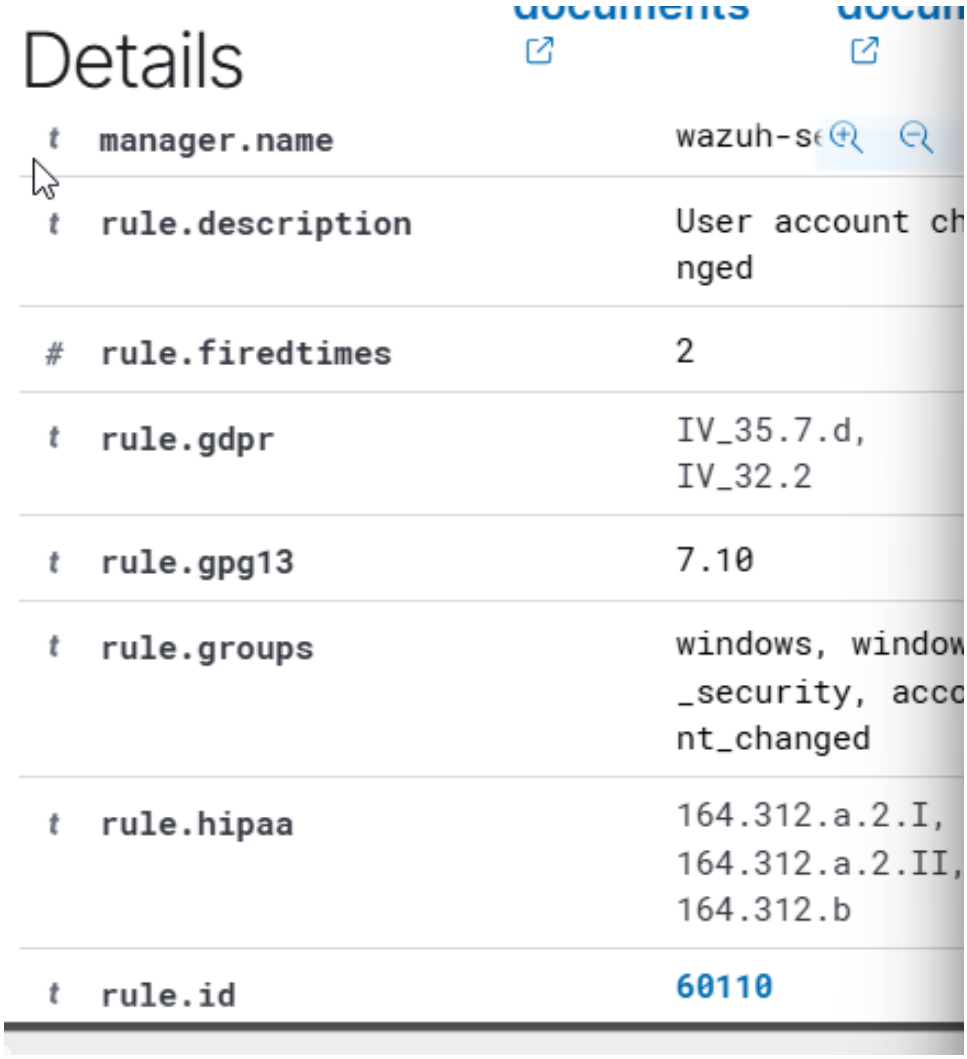
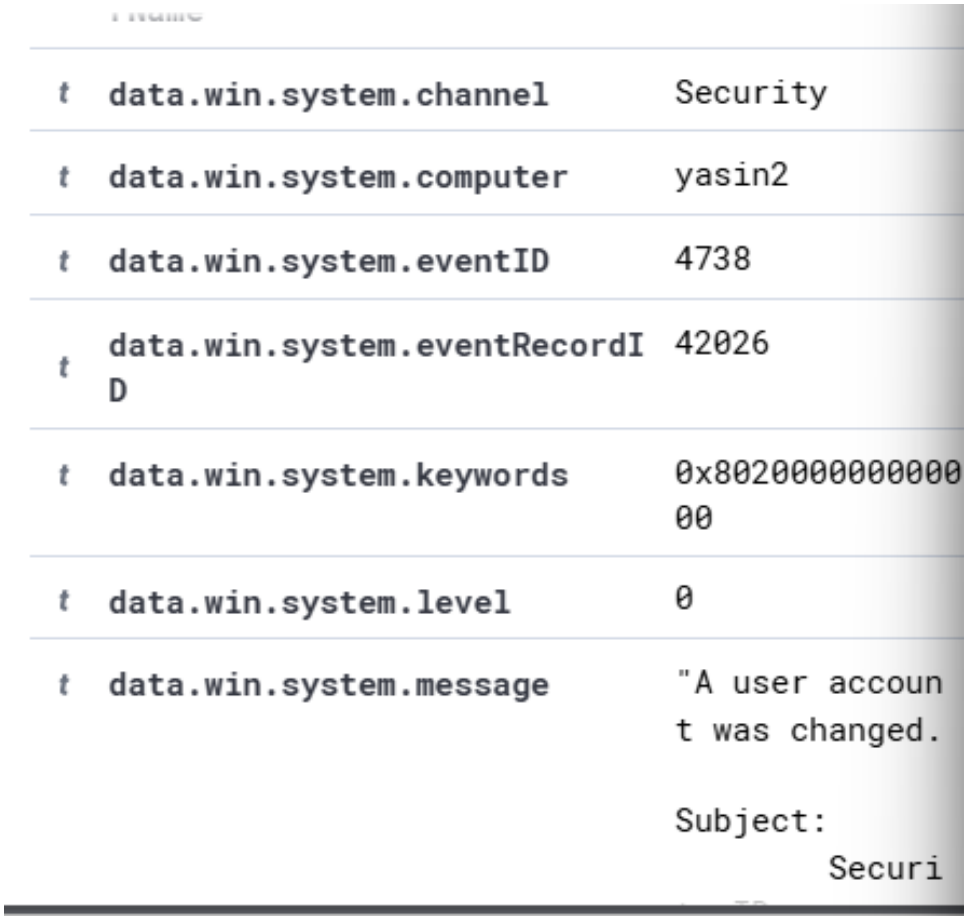


Figure 8. Wazuh rule details for the user-account-changed event.



t	data.win.system.channel	Security
t	data.win.system.computer	yasin2
t	data.win.system.eventID	4738
t	data.win.system.eventRecordID	42026
t	data.win.system.keywords	0x8020000000000000
t	data.win.system.level	0
t	data.win.system.message	"A user account was changed. Subject: Securi

Figure 9. Windows Security Event ID 4738 confirmation.

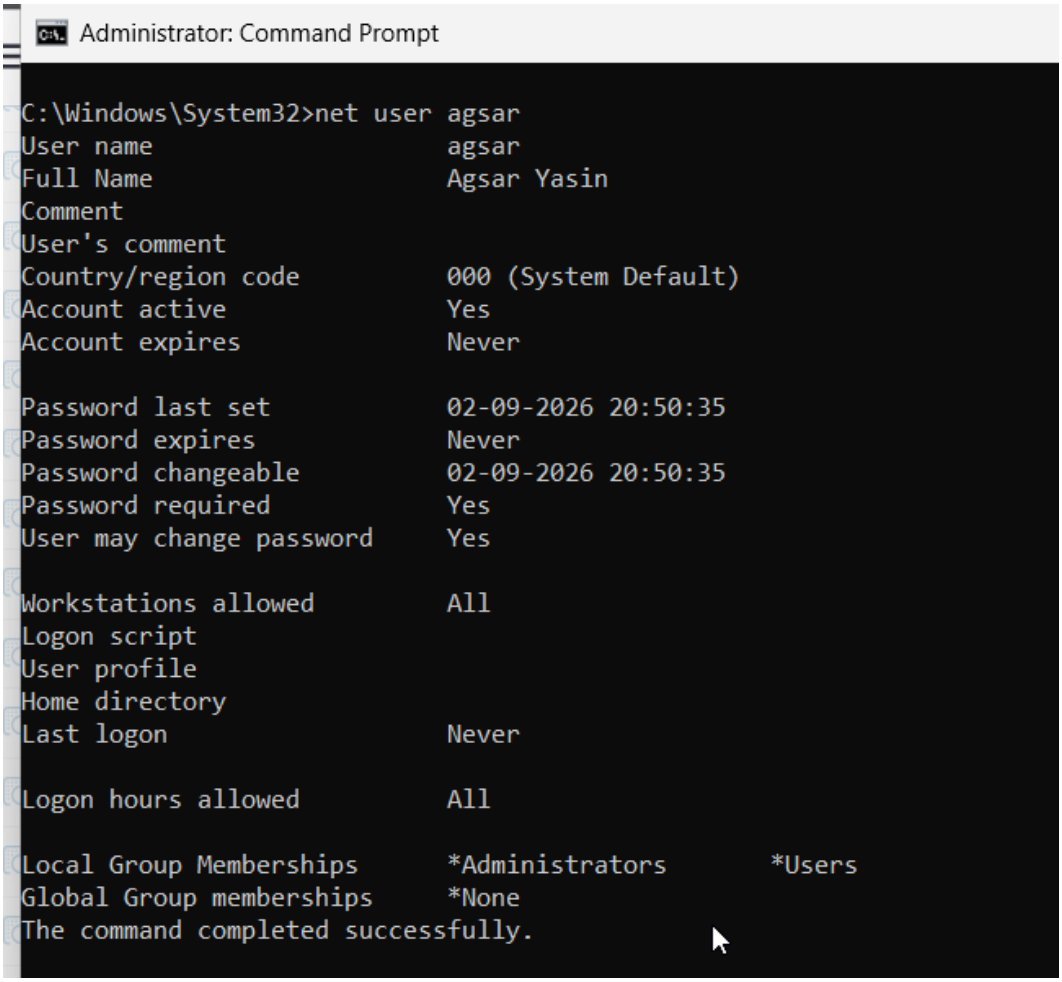


Figure 10. Windows net user agsar verification used to check the account state.

3.7 Scenario 1 lessons

- Event ID 4625 is the key Windows evidence for a failed logon.
- The analyst should verify the account, source, logon type, process, time and surrounding events.
- A single failed logon is not automatically malicious.
- Unexpected nearby events should be investigated separately rather than assuming causation.

4. Scenario 2 — Multiple Failed Logins / Repeated Authentication Failures

Objective: demonstrate how repeated authentication failures can be identified in a SIEM and investigated as a possible credential-attack pattern.

4.1 Controlled test

Four visible Windows 4625 events were generated in a short period. The events occurred at 17:14:44.093, 17:14:49.699, 17:14:54.063 and 17:14:58.067 on 8 September 2026. All four were associated with yasin2 and were shown as Logon Failure events at rule level 5.

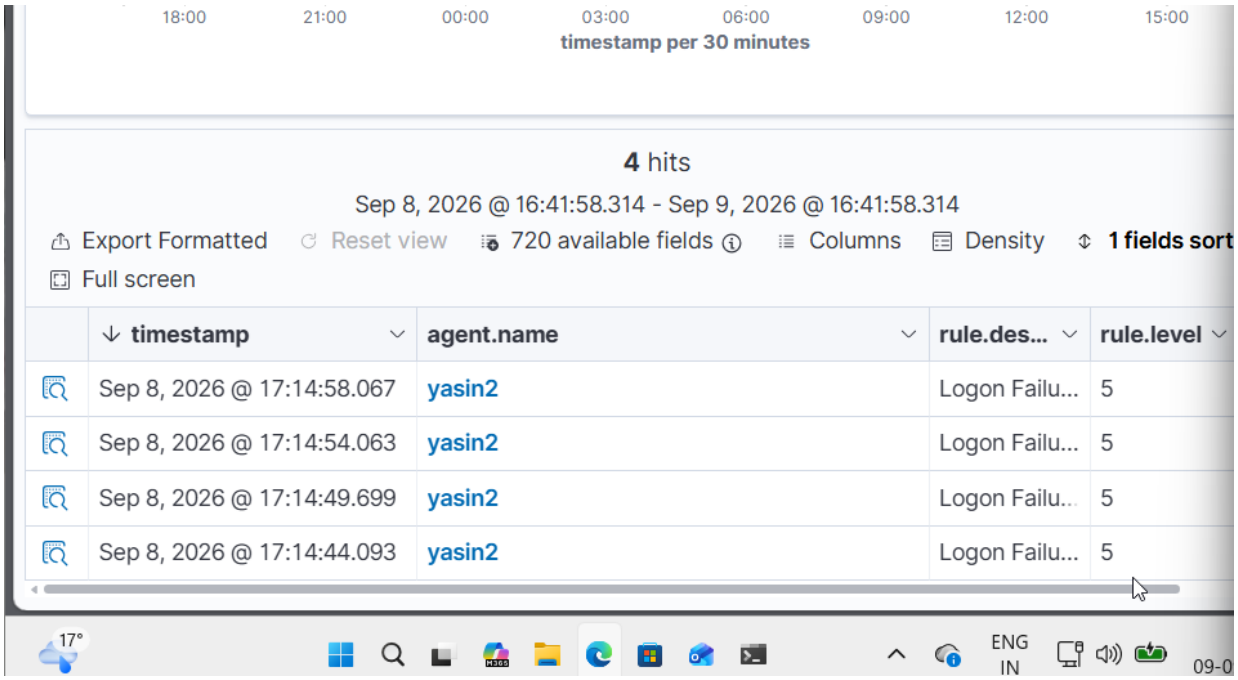


Figure 11. Wazuh Events view showing four repeated Logon Failure events for yasin2.

4.2 Investigation

The investigation searched for Event ID 4625 and reviewed the repeated timestamps. One event showed rule.firedtimes: 4, consistent with the repeated test activity. A follow-up search excluding rule 60122 returned no results, so no separate higher-level brute-force correlation rule was observed during this test.

Investigation item	Result
Windows event	4625
Number of visible events	4
Time window	Approximately 15 seconds
Endpoint	yasin2 / Agent 002
Rule observed	60122, level 5
Higher-level brute-force rule	Not observed in the tested search
Classification	Controlled repeated authentication activity
Response	No containment; test activity was known and controlled.

4.3 SOC interpretation

Repeated failed logins are a useful detection signal because they can occur during password guessing or other credential attacks. In a real environment, the analyst would correlate the frequency, source address, target accounts, successful logons, geolocation where appropriate, endpoint role, and other surrounding events before deciding whether escalation or containment is justified.

4.4 Scenario 2 lessons

- Repeated Event ID 4625 events provide stronger context than a single failure.
- Frequency and timing are important investigation dimensions.
- Do not claim a brute-force attack solely from repeated failures; establish the evidence and context.
- Correlation rules can add value, but the analyst must verify which rule actually fired.

5. Scenario 3 — Suspicious Process Detection

Objective: detect Windows process creation, create a custom Wazuh rule, verify the rule with wazuh-logtest, generate live alerts, and investigate the resulting event.

5.1 Enable process-creation auditing

Windows Process Creation auditing was enabled so that process starts generate Security Event ID 4688.

```
auditpol /set /subcategory:"Process Creation" /success:enable

auditpol /get /subcategory:"Process Creation"
```

The audit policy was verified as enabled. Windows subsequently generated multiple Event ID 4688 records.

5.2 Confirm Windows Event 4688

Get-WinEvent -LogName Security -MaxEvents 100 | Where-Object {$_.Id -eq 4688} | Select-Object -First 3 TimeCreated, Id, ProviderName

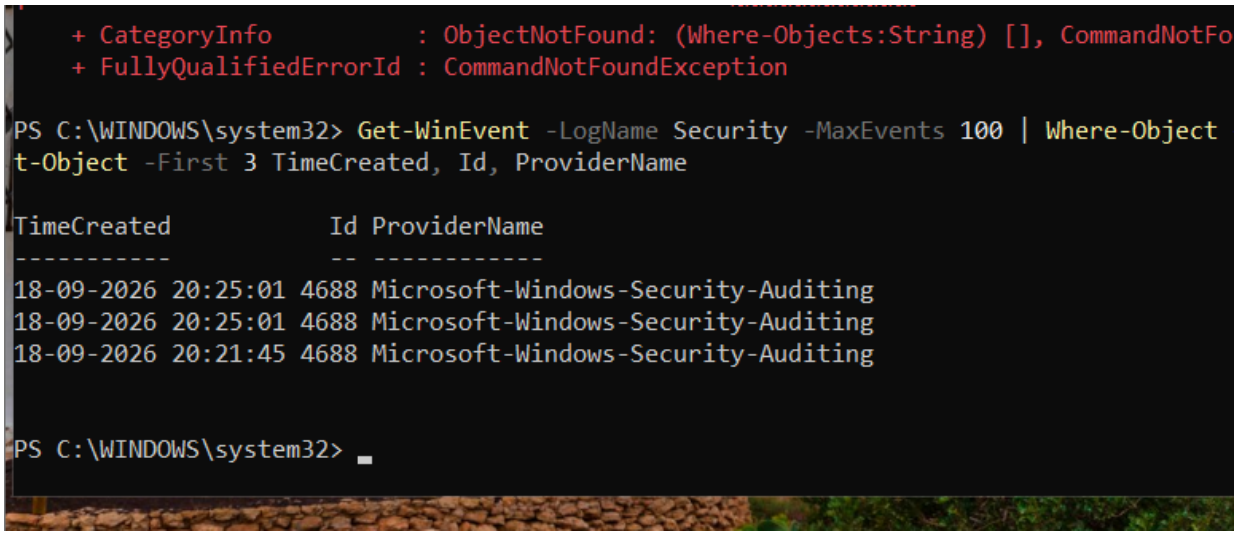


Figure 12. Windows PowerShell confirming recent Event ID 4688 records from Microsoft-Windows-Security-Auditing.

5.3 Wazuh agent and network verification

The WazuhSvc service was confirmed Running. The agent configuration contained the Wazuh manager address 192.168.1.104 and a Security EventChannel configuration. TCP connectivity to port 1514 was confirmed successfully.

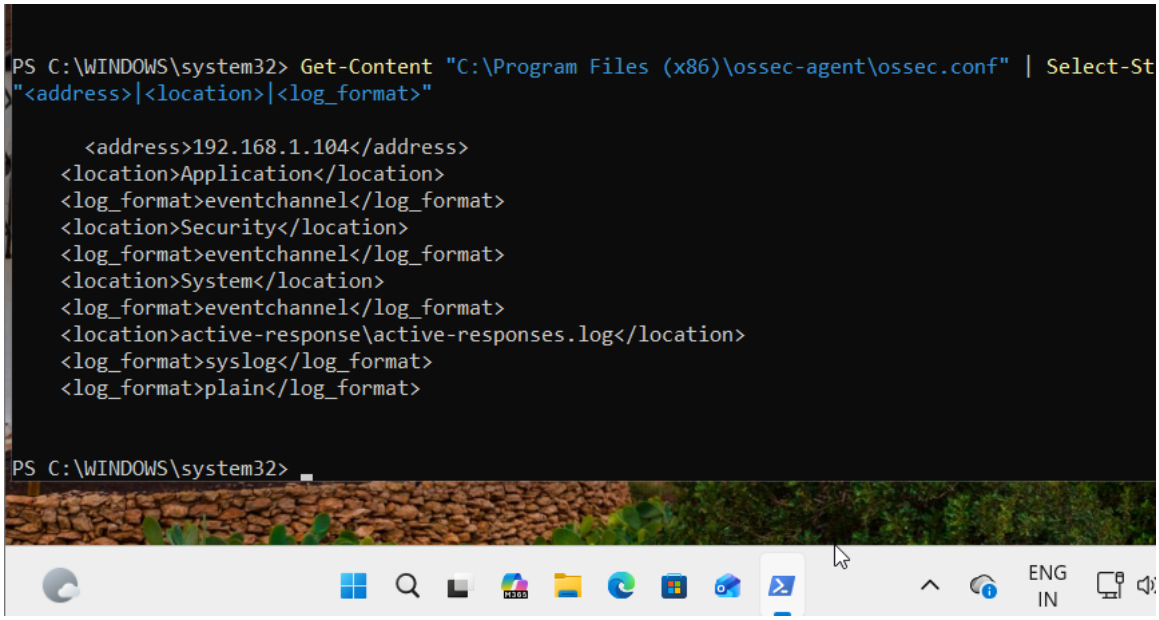


Figure 13. Wazuh agent configuration showing manager address 192.168.1.104 and Security EventChannel monitoring.

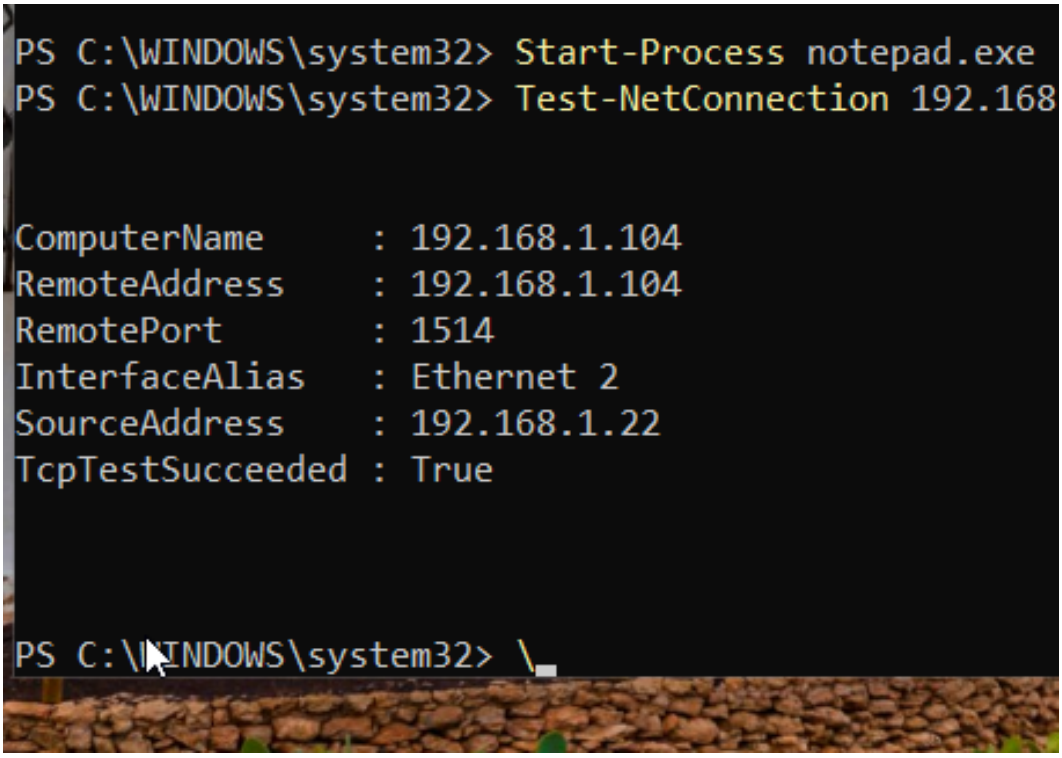


Figure 14. Windows Test-NetConnection confirming TCP 1514 connectivity to the Wazuh server.

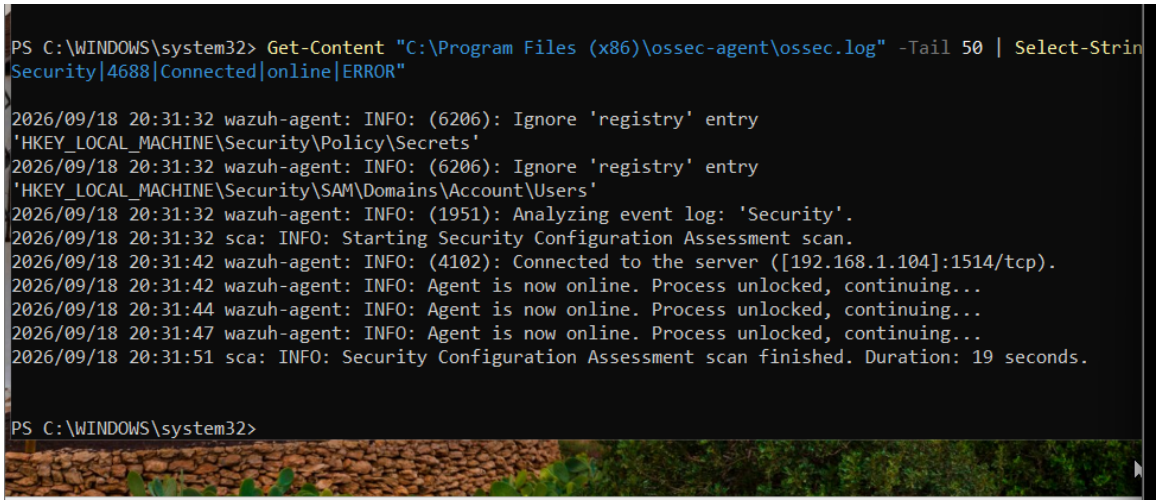


Figure 15. Wazuh agent log showing Security log analysis and a successful connection to 192.168.1.104:1514.

5.4 Built-in Wazuh rule 67027

The built-in Wazuh process-creation rule was verified as rule 67027. It matches the Windows Security provider and Event ID 4688 and has the description 'A process was created.'

This verification was important because an earlier rule ID, 67007, was initially considered but was found to belong to Windows Firewall rules rather than process creation. The correct built-in process-creation rule was 67027.

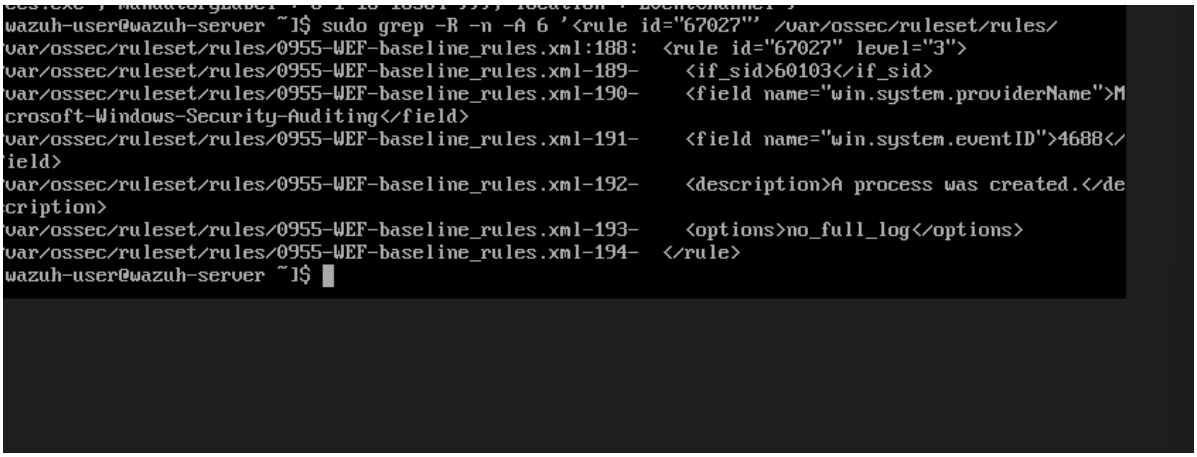


Figure 16. Wazuh ruleset search confirming rule 67027 and its Event ID 4688 conditions.

5.5 Custom Wazuh rule 100100

A custom Level 5 rule was created in local_rules.xml to detect Windows Event ID 4688 directly:

```
<group name="windows,process_creation,">
  <rule id="100100" level="5">
    <field name="win.system.eventID">4688</field>
    <description>Windows Process Creation Detected</description>
    <group>process_creation,</group>
  </rule>
</group>
```



Figure 17. Terminal verification showing custom rule 100100 saved with Event ID 4688 and Level 5.

5.6 Rule testing with wazuh-logtest

The custom rule was tested with a simplified JSON representation of Event ID 4688. The test completed all three phases and matched rule 100100.

```
printf '%s\n' '{"win":{"system":{"providerName":"Microsoft-Windows-Security-Auditing","eventID":"4688"}}}' | sudo
/var/ossec/bin/wazuh-logtest
```

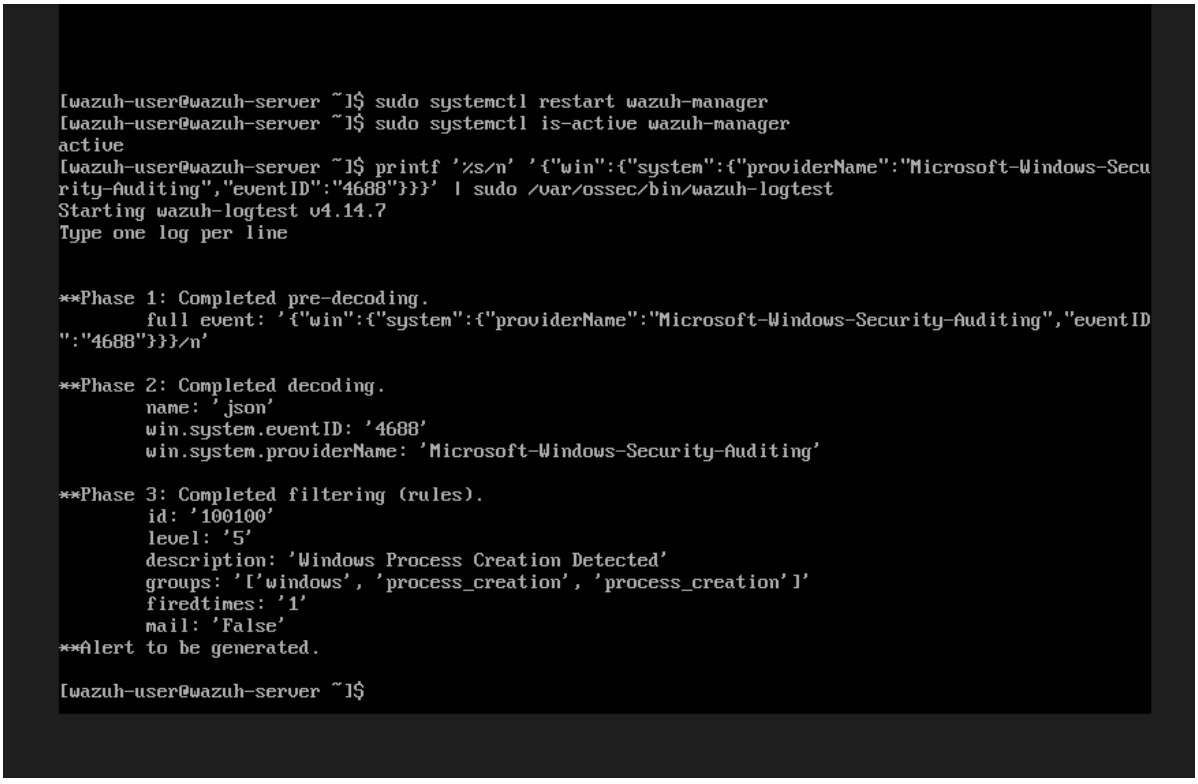


Figure 18. wazuh-logtest confirming Phase 3 and custom rule 100100, Level 5.

5.7 Live Wazuh alert

After the rule was validated, live process-creation events were searched in the Wazuh Dashboard using rule.id:100100. The dashboard displayed hundreds of matching events from yasin2, demonstrating that the custom rule was working in the live pipeline.

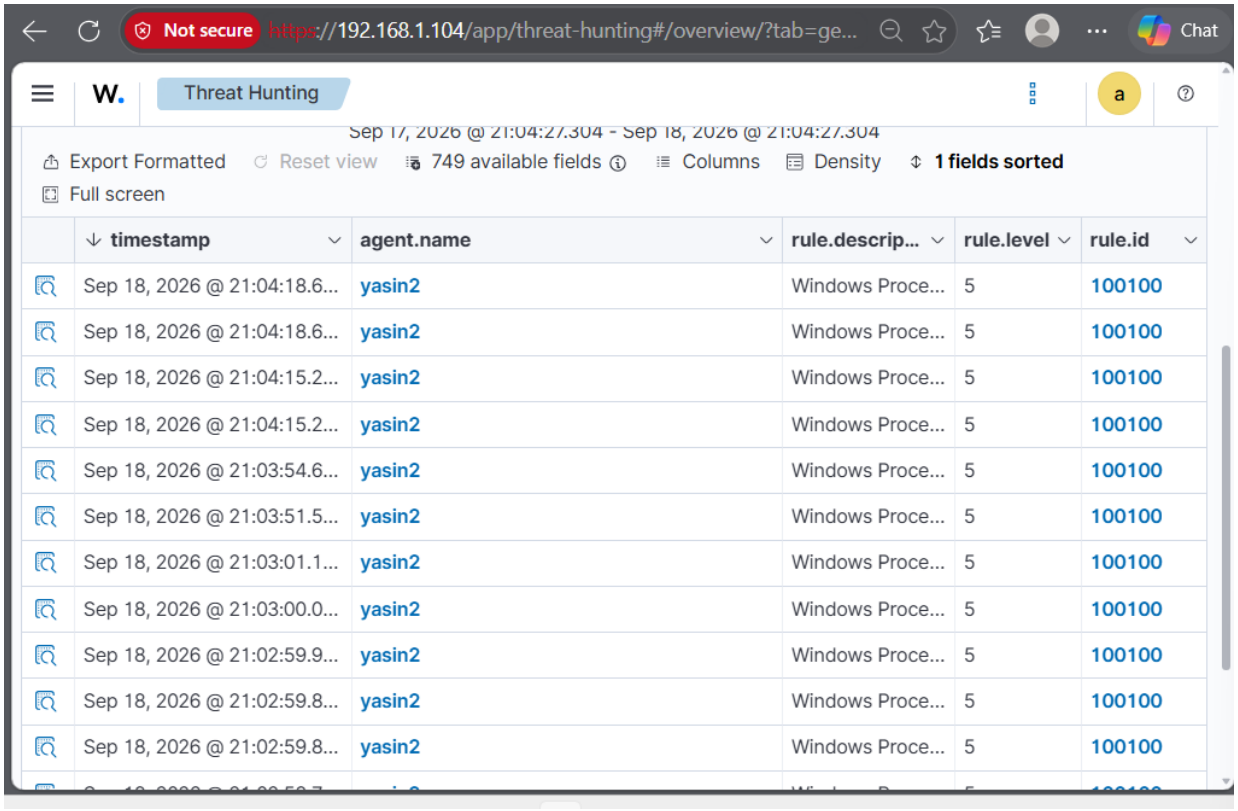


Figure 19. Wazuh Threat Hunting view showing live rule 100100 alerts for yasin2.

5.8 Investigation of a live alert

One live alert was opened for detailed investigation. The event details confirmed Agent ID 002, agent name yasin2, current endpoint IP 192.168.1.22, Windows Security channel, Event ID 4688, Microsoft-Windows-Security-Auditing as the provider, and custom rule 100100 at Level 5.

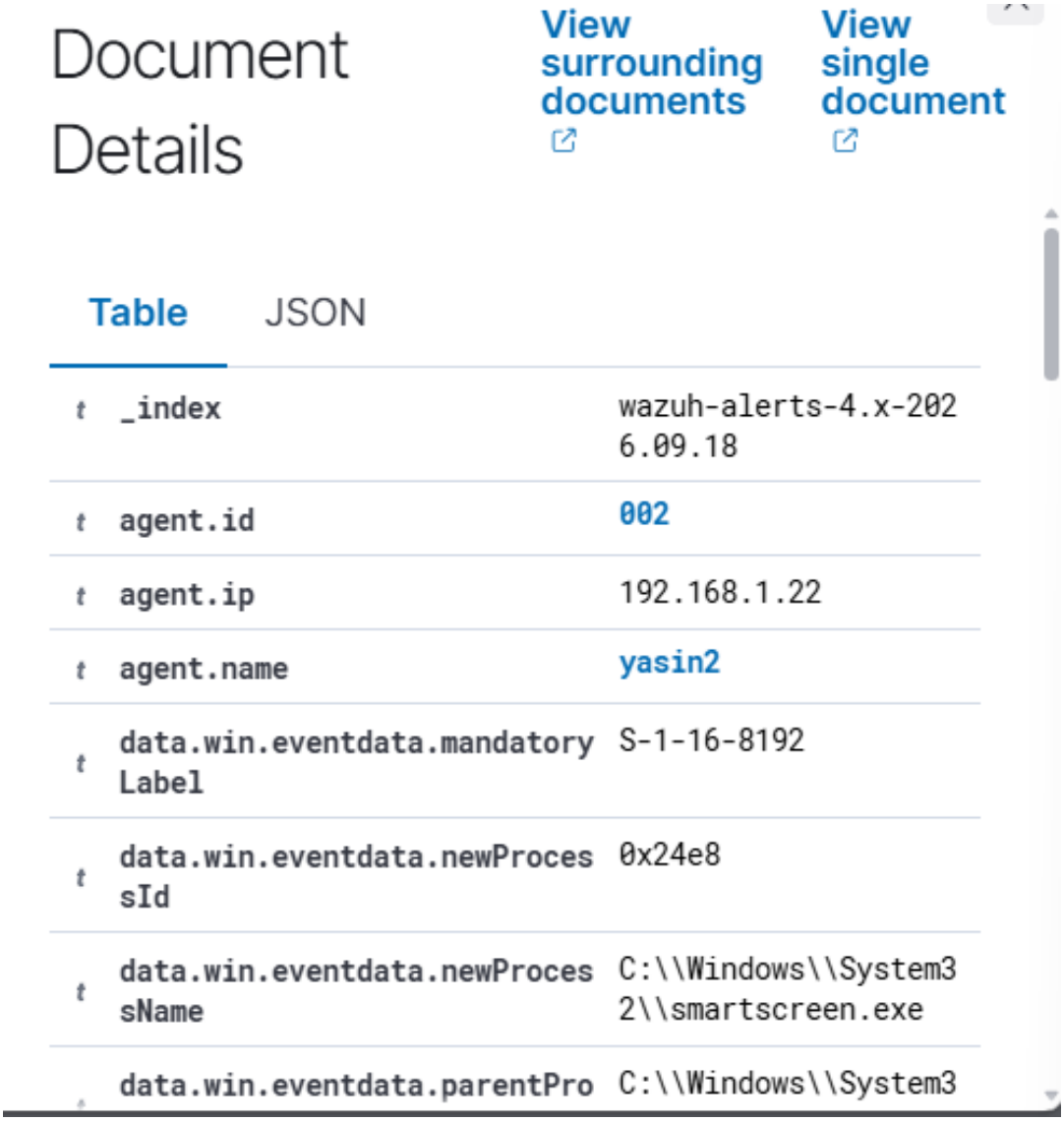


Figure 20. Expanded Wazuh alert showing Agent 002, yasin2, endpoint IP 192.168.1.22 and the new process path.

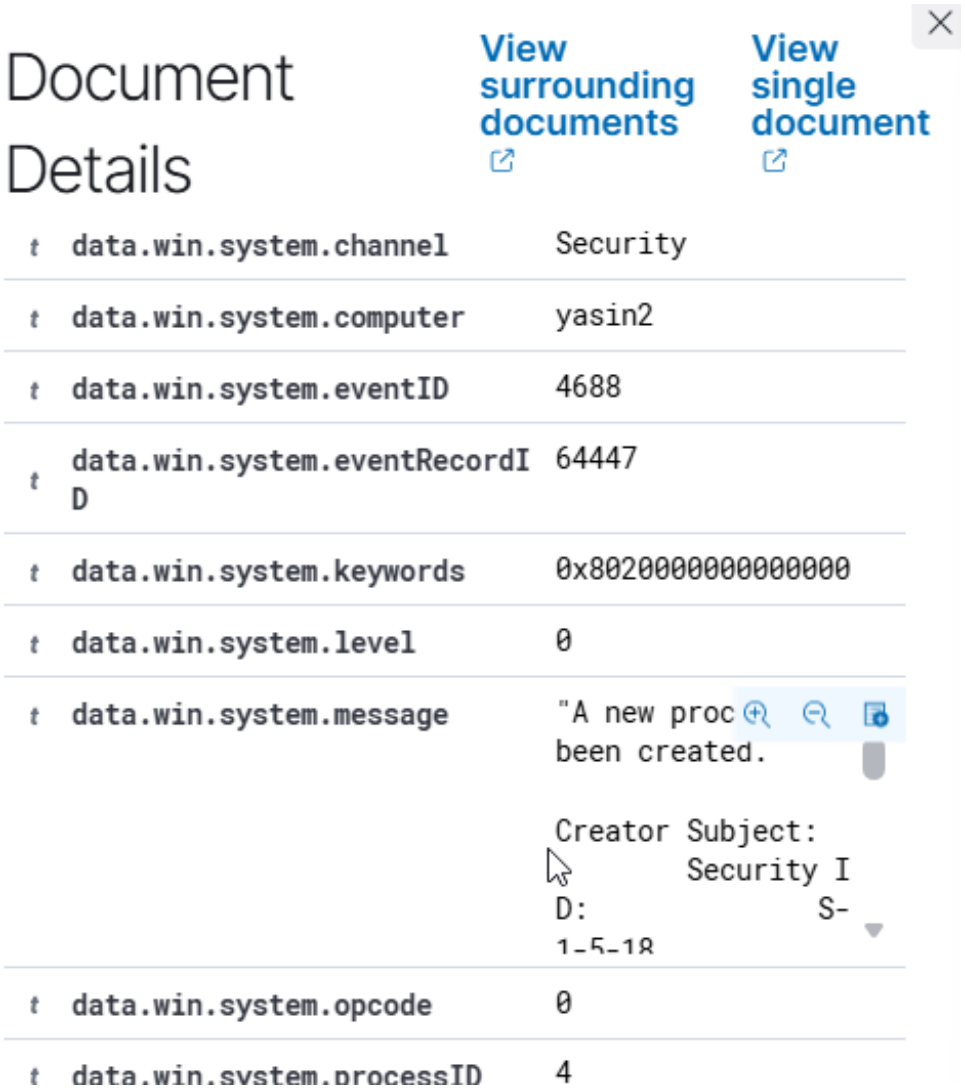


Figure 21. Event details showing Event ID 4688, Security channel and Microsoft-Windows-Security-Auditing provider.

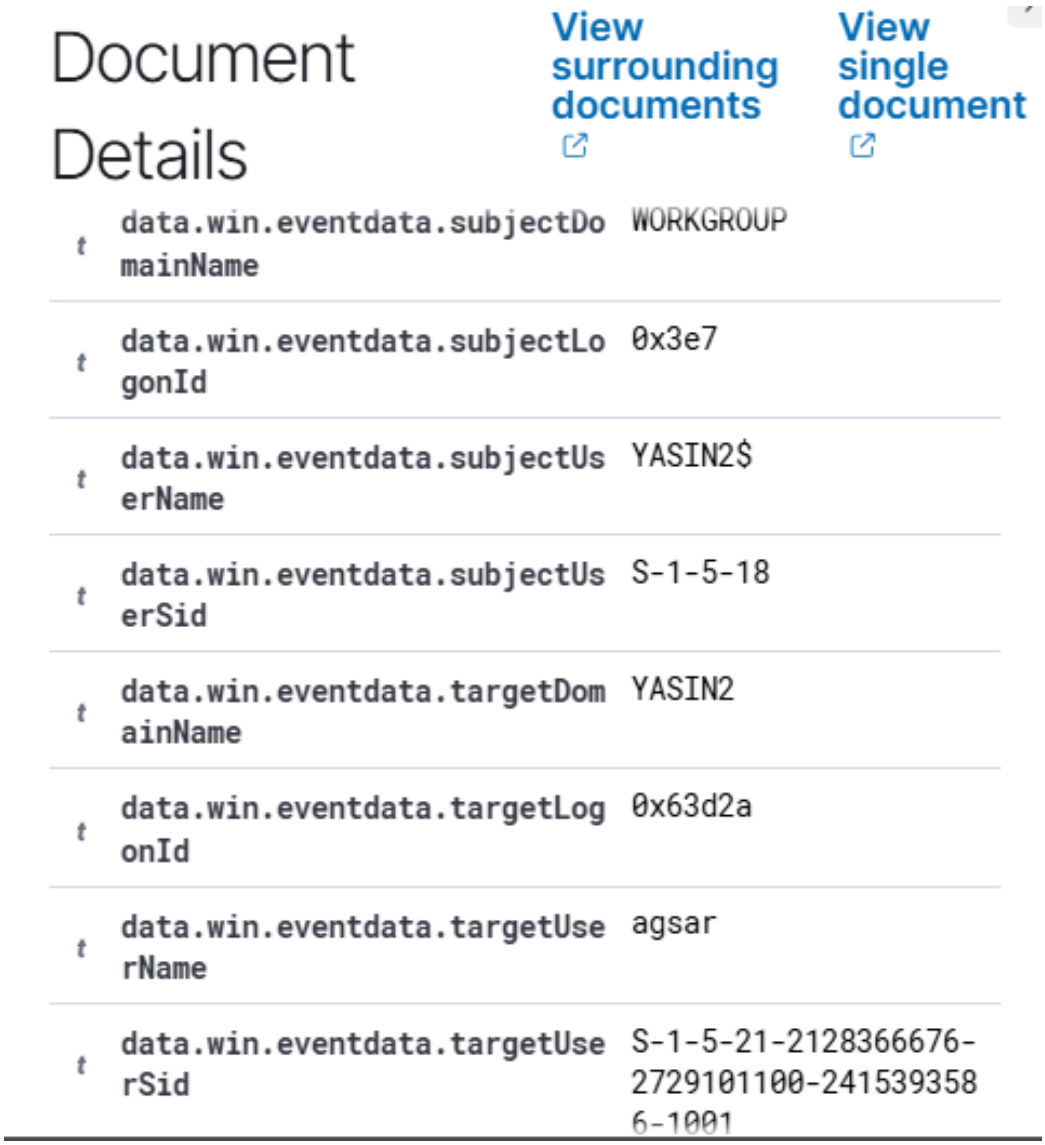


Figure 22. Event details showing the subject and target account fields for the process-creation event.

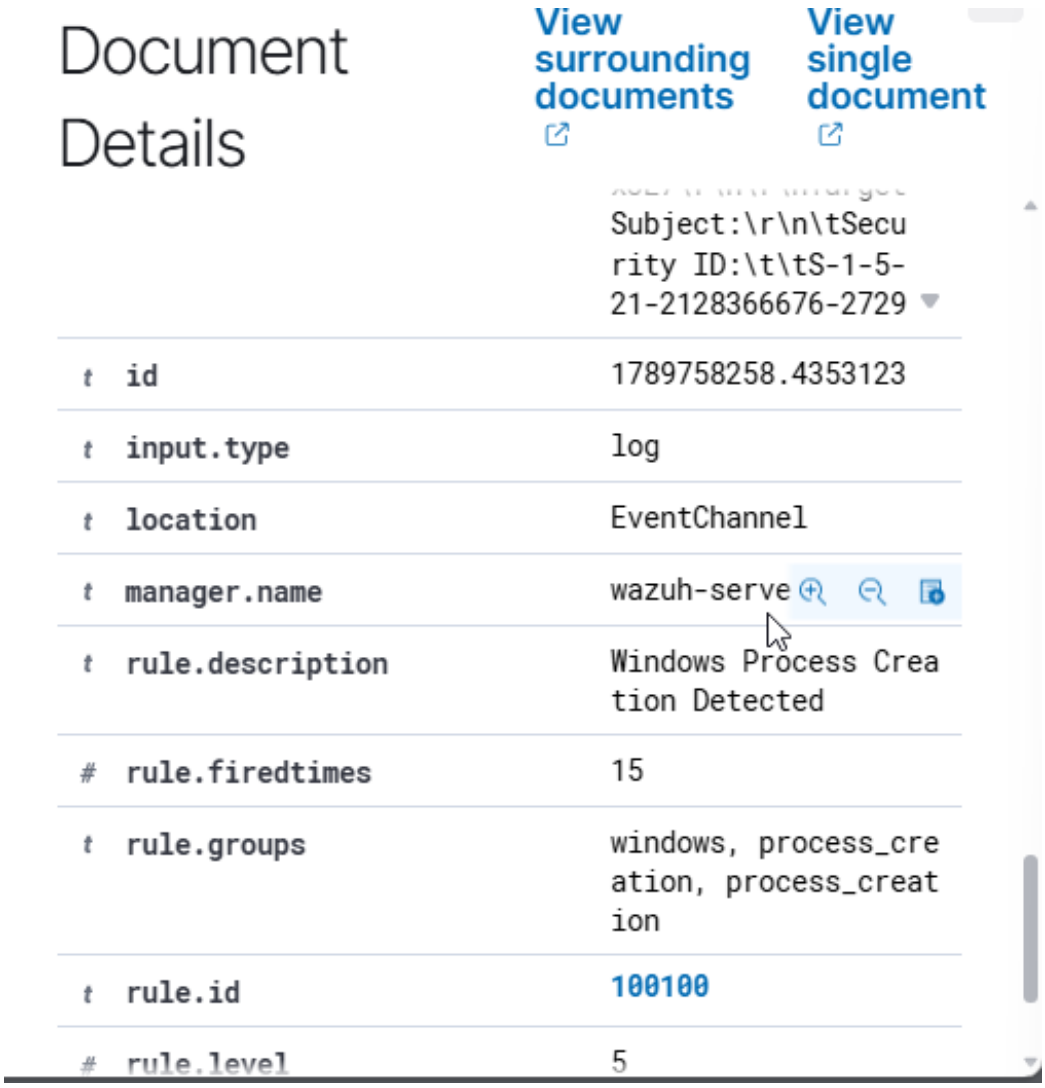


Figure 23. Event details confirming custom rule 100100, description and Level 5.

5.9 Classification and response

Item	Finding
Event ID	4688 — new process created
Built-in Wazuh rule	67027 — A process was created.
Custom Wazuh rule	100100 — Windows Process Creation Detected
Custom severity	Level 5
Agent	002 / yasin2
Endpoint IP	192.168.1.22
Observed process	C:\Windows\System32\smartscreen.exe
Provider	Microsoft-Windows-Security-Auditing
Channel	Security
Classification	Benign/expected Windows system activity in the controlled lab context
Response	No containment or eradication required; alert investigated and documented.

5.10 Scenario 3 lessons

- Event ID 4688 is a core Windows process-execution event for SOC monitoring.
- Wazuh's built-in rule 67027 can identify process creation, while a custom rule can provide a project-specific alert.
- wazuh-logtest is useful for validating custom rules before relying on live events.

- An analyst should inspect process path, command line, user, parent process, process ID and surrounding events.
- A process-creation alert does not automatically mean malware; context determines classification.

6. Cross-Scenario SOC Analysis

The three completed scenarios demonstrate increasing levels of investigation complexity: a single authentication failure, repeated authentication failures, and process creation. Together they show how a SOC analyst moves from raw Windows telemetry to an actionable security decision.

SOC stage	Scenario 1	Scenario 2	Scenario 3
Trigger	Wrong password	Repeated wrong passwords	Process start
Windows event	4625	4625	4688
Primary question	Who failed to log in and from where?	Is the repeated pattern suspicious?	What process was created and by whom?
Investigation	Account, source, logon type, status	Frequency, timing, repetition, rule behaviour	Process path, parent, user, command line, event context
Classification	Controlled/benign	Controlled repeated activity	Benign/expected system activity
Response	Document	Document; no containment	Document; no containment

SOC investigation mental model

SOC question → Windows activity → log source → Event ID → SIEM/Wazuh search → evidence → classification → response → documentation

7. Commands and Tools Used

Command/tool	Purpose	Study priority
Get-WinEvent	Read Windows event logs	Memorise purpose
auditpol	Check/enable Windows audit policy	Understand
Get-Service WazuhSvc	Check Wazuh Windows service	Recognise
Restart-Service WazuhSvc	Restart Wazuh Windows agent	Recognise
Test-NetConnection	Test TCP connectivity to Wazuh	Memorise purpose
Get-Content ossec.log	Read Wazuh agent troubleshooting log	Recognise
grep	Search Linux text/log files	Memorise basic use
tail	View recent Linux log lines	Memorise basic use
systemctl	Manage/check Linux services	Memorise basic use
wazuh-logtest	Test Wazuh rules	Memorise purpose
Nmap	Network discovery and service enumeration	Know purpose for next project
Autopsy	Digital forensic examination	Know purpose for next project
FTK Imager	Forensic imaging/evidence acquisition	Know purpose for next project
Wireshark	Network traffic capture/analysis	Know purpose

8. What to Memorise for SOC Interviews

- 4624 = successful Windows logon.
- 4625 = failed Windows logon.
- 4688 = new process created.
- 4738 = user account changed.
- CIA = Confidentiality, Integrity, Availability.
- SIEM = central collection/search/investigation of security events.
- Wazuh Agent = endpoint data collection/forwarding.
- Wazuh Manager = receives events, decodes them and applies rules.
- Event = recorded activity; alert = event/pattern that matched a detection rule.
- MITRE ATT&CK = knowledge base of adversary tactics and techniques.
- TCP 1514 is the Wazuh agent communication port used in this lab.
- Nmap = network/service discovery and enumeration.

- Autopsy = digital forensic analysis platform.
- FTK Imager = forensic acquisition/imaging tool.

9. Lessons Learned

- Troubleshooting should follow the data path rather than changing several controls at once.
- The Windows event must exist before the SIEM can detect it.
- The endpoint must be connected and correctly configured before events can reach the manager.
- Rule testing with wazuh-logtest can isolate detection-rule problems from collection problems.
- Built-in rule IDs must be verified before chaining custom rules to them.
- A good SOC report records evidence, not just conclusions.
- Controlled benign events are useful for demonstrating detection engineering and investigation skills.
- The same investigation discipline can be applied to real alerts, with additional correlation and escalation where required.

10. Final Portfolio Evidence Checklist

- ✓ Scenario 1: Dashboard authentication-failure evidence.
- ✓ Scenario 1: Agent 002 filtering evidence.
- ✓ Scenario 1: Event ID 4625 event evidence.
- ✓ Scenario 1: Authentication details including Logon Type 2 and status codes.
- ✓ Scenario 1: Additional Event ID 4738 investigation and account verification.
- ✓ Scenario 2: Four repeated Event ID 4625 / Logon Failure events visible in Wazuh.
- ✓ Scenario 3: Windows Event ID 4688 confirmation.
- ✓ Scenario 3: Wazuh agent configuration and Security EventChannel evidence.
- ✓ Scenario 3: TCP 1514 connectivity evidence.
- ✓ Scenario 3: Built-in rule 67027 verification.
- ✓ Scenario 3: Custom rule 100100 configuration evidence.
- ✓ Scenario 3: wazuh-logtest Phase 3 match to rule 100100.
- ✓ Scenario 3: Live Dashboard rule 100100 alerts.
- ✓ Scenario 3: Expanded alert with Agent 002/yasin2, Event ID 4688 and process details.

Final status

SCENARIO 1 — COMPLETE | SCENARIO 2 — COMPLETE | SCENARIO 3 — COMPLETE

Next planned portfolio projects: Vulnerability Assessment (Nmap + OpenVAS/Greenbone), Digital Forensics Investigation (Autopsy + FTK Imager), and Phishing Email Investigation.

Appendix A — Evidence Screenshot Index

Figure	Scenario	Evidence
1–10	Scenario 1	Authentication failure, endpoint filtering, Event 4625 investigation, Event 4738 verification
11	Scenario 2	Four repeated Logon Failure events for yasin2
12	Scenario 3	Windows Event 4688 confirmation
13–15	Scenario 3	Agent configuration, network connectivity and agent log
16	Scenario 3	Built-in Wazuh rule 67027
17	Scenario 3	Custom Wazuh rule 100100
18	Scenario 3	wazuh-logtest successful rule match
19	Scenario 3	Live Wazuh rule 100100 results
20–23	Scenario 3	Expanded live alert and process-creation investigation

Screenshot selection principle: screenshots are included where they provide meaningful evidence for detection, investigation, validation, or classification. Routine troubleshooting screens that did not add portfolio evidence were intentionally not included.